

AI-Based Farmer Query Support and Advisory System

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Abstract—The AI-Based Farmer Query Support and Advisory System is a web-based platform designed to assist farmers by leveraging artificial intelligence and real-time data to improve agricultural decision-making. Traditional farming advisory methods often lack real-time, context-aware support. This system addresses this gap by enabling users to submit queries and receive personalized responses generated by a lightweight large language model (Gemma 3.1B via Ollama). Furthermore, it integrates a deep learning-based module utilizing MobileNetV2 for accurate plant disease detection from uploaded crop images. Real-time weather data is also incorporated to ensure recommendations are location-specific. The system is built on a scalable FastAPI backend, a MySQL database, and a responsive HTML/CSS/JavaScript frontend. By combining natural language processing and computer vision, this platform enhances access to vital agricultural knowledge and promotes informed modern farming practices.

Index Terms—smart farming, artificial intelligence, crop disease detection, large language models, e-agriculture.

I. INTRODUCTION

Agriculture plays a fundamental role in sustaining economies and livelihoods, yet farmers frequently encounter modern challenges such as unpredictable weather, pest outbreaks, and crop diseases. Traditional advisory systems—relying on manual expertise, extension services, and static online resources—are often insufficient, lacking the capacity for real-time interaction and personalization. To address these limitations, this paper proposes an AI-Based Farmer Advisory System. The system provides an accessible web interface where farmers can receive expert-level guidance, diagnose plant diseases via image analysis, and obtain context-aware recommendations driven by real-time environmental factors. By bridging the gap between advanced AI technologies and practical farming applications, the system aims to reduce dependency on manual consultations and enhance overall crop productivity.

II. LITERATURE REVIEW

The integration of Artificial Intelligence into agriculture has been widely explored to meet global food security demands. Liu (2020) highlighted the necessity of AI in transitioning to Agriculture 5.0, utilizing machine learning for data-driven decision-making in crop rotation and disease detection. Elbasi et al. (2023) proposed an integrated framework combining AI and the Internet of Things (IoT) for predictive analytics and real-time soil monitoring, demonstrating high accuracy in plant disease detection using classification algorithms. Addressing the "black box" nature of AI, Holzinger et al. (2024) emphasized Human-Centered AI (HCAI), integrating Explainable AI (XAI) to foster trust and collaborate with human expertise rather than replace it. Similarly, Rani et al. (2023) explored a data-driven smart farming architecture that combines satellite imagery and sensor data to predict environmental stress, allowing for timely preventive measures. Building upon these foundations, the proposed system implements real-time, context-aware AI interactions and computer vision in a lightweight, accessible web application.

III. PROPOSED SYSTEM

A. System Architecture

The proposed system follows a modular, three-tier client-server architecture:

- **Frontend Layer:** A responsive web interface built using HTML, CSS, and JavaScript, enabling farmers to submit queries and upload images.
- **Backend Layer:** A high-performance FastAPI server handles routing, request processing, and RESTful API communication. It interacts with external APIs for weather data.
- **Database & AI Layer:** A MySQL database manages user profiles, chat histories, and feedback. The AI core utilizes the Ollama framework hosting Gemma 3.1B

for natural language processing, and TensorFlow/Keras running MobileNetV2 for image classification.

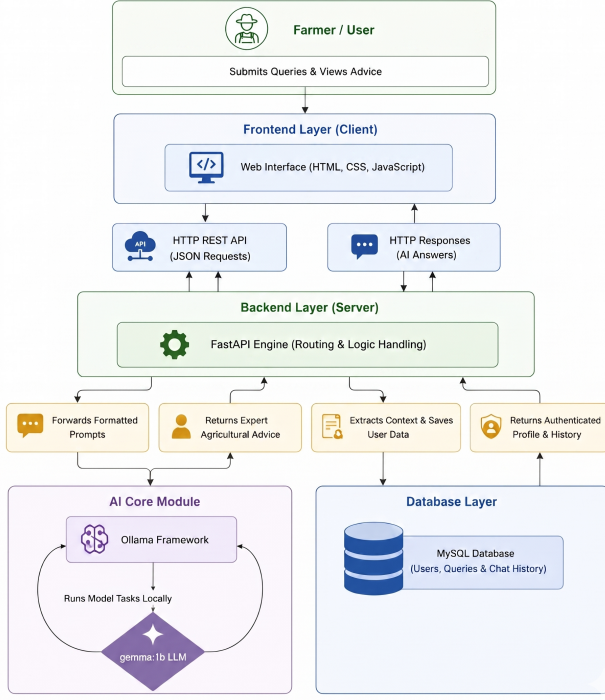


Fig. 1. System Architecture Diagram

B. Functional Modules

- 1) **User Authentication:** Users securely register and log in, enabling personalized session tracking and location-based services.
- 2) **AI Query Processing:** Text queries are transformed into structured prompts and processed by the Gemma 3.1B model to generate context-aware agricultural advice.
- 3) **Plant Disease Detection:** Farmers upload images of affected crops. The system preprocesses the image and passes it through a pre-trained MobileNetV2 model to classify and predict the disease.
- 4) **Weather Integration:** The backend fetches real-time environmental data based on the user's stored location using external weather APIs, providing necessary context for the LLM's recommendations.

C. Data Flow

The system's data flow is designed for rapid processing and persistence. In the Level 0 Context Diagram, the external entities (Farmer and Admin) interact centrally with the Advisory System. The Level 1 Data Flow details the internal routing: user credentials validate against the database (D1); agricultural queries are structured and sent to the external LLM; and the generated formatted advice is returned to the user while being simultaneously archived in the Query History Database (D2).

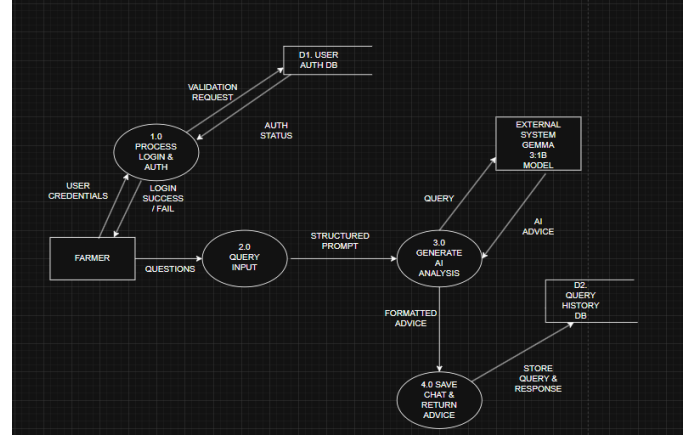


Fig. 2. Data Flow Diagram (Level 1)

D. Algorithm

The core execution flow of the system is as follows:

- 1) User authenticates and submits a text query or image.
- 2) Backend validates the incoming HTTP request.
- 3) If an image is present, it is preprocessed and passed to MobileNetV2 to extract the disease class prediction.
- 4) The user's location is queried from the database to fetch real-time weather data via API.
- 5) The text query, predicted disease (if any), and weather context are concatenated into a structured prompt.
- 6) The prompt is fed to the Gemma 3.1B LLM to generate actionable advice.
- 7) The query and response are saved to the database.
- 8) The formatted response is returned to the frontend interface.

IV. RESULTS AND DISCUSSION

A. Evaluation Metrics

The system underwent rigorous unit, integration, and system testing to validate functional and non-functional requirements. Table I summarizes the expected versus obtained outcomes during system evaluation.

TABLE I
SYSTEM EVALUATION METRICS

Feature Tested	Test Description	Expected Result	Actual	Status
User Authentication	Verify complete login process	Secure dashboard access	Success	Pass
AI Query System	Submit complex agricultural query	Relevant, context-aware advice	Success	Pass
Disease Detection	Upload infected crop image	Correct disease classification	Success	Pass
Weather Integration	Fetch data based on user location	Real-time weather displayed	Success	Pass

B. Discussion

The integration of MobileNetV2 demonstrated satisfactory accuracy in classifying multiple crop diseases, proving reliable

for early diagnosis. The use of the FastAPI framework successfully managed asynchronous requests, maintaining low-latency responses for both text processing and image inference. By routing queries through the Gemma 3.1B model, the system consistently generated concise, context-aware recommendations, significantly improving upon static agricultural databases. The user interface maintained high accessibility, allowing users with minimal technical expertise to navigate the platform seamlessly.

V. LIMITATIONS

While effective, the current system presents certain limitations:

- **Connectivity:** The architecture relies on stable internet access to reach weather API, posing challenges for remote rural areas.
- **Model Constraints:** The disease detection model's accuracy is bound by its training dataset and may struggle with rare or unseen plant diseases.
- **Security:** Authentication relies on standard credential matching; it lacks advanced encryption or token-based multi-factor authentication.
- **Accessibility:** The lack of native multilingual support restricts usability for farmers who rely strictly on local regional languages.

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VII. CONCLUSION AND FUTURE SCOPE

The AI-Based Farmer Query Support and Advisory System successfully bridges the gap between complex artificial intelligence and practical, everyday farming needs. By integrating the Gemma 3.1B language model for natural language processing and MobileNetV2 for image-based disease classification, the platform empowers farmers to make data-driven, timely decisions without relying solely on manual expert consultation. Future enhancements will focus on expanding the image training datasets for broader disease coverage and integrating robust multilingual support to improve rural accessibility. Furthermore, implementing voice-to-text functionalities and developing a dedicated mobile application with offline capabilities will significantly increase the system's operational flexibility. Finally, integrating IoT field sensors for real-time soil and moisture analytics will transition the platform toward a fully automated precision farming ecosystem.

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